

Shijie Xiao

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EDUCATION

Georgia Institute of Technology, Atlanta, GA

May 2026

Master of Science in Computer Engineering

Relevant Coursework: **Computer Vision, Adv Prog Techniques**, Generate and Geometric DL, ML for Neural/Behavioral Data

Huazhong University of Science and Technology, Wuhan, China

June 2024

Bachelor of Science in Artificial Intelligence & Automation

GPA: **3.95/4.00**

Relevant Coursework: **Data Structures and Algorithms, Machine Learning, Principles of Computer Composition**, Linear Algebra, Calculus, Probability and Statistics

SKILLS

Programming Python, C++&C, Java, Bash, MATLAB, Linux **AI Frameworks** PyTorch, Tensorflow, OpenCV

Language Mandarin (Native), English **Version Control** Git, Excel, SQL **Model** Diffusion Model, Transformer, CNN, LSTM

PROJECTS

Camera-based Subway Congestion Prevention System | PyTorch, OpenCV, ST-GCN, YOLOv8, ByteTrack Feb-June 2023

- Improved ByteTrack algorithm, achieving accurate multi-person parallel recognition despite occlusions.
- Trained YOLOv8 detector on MOT dataset, enhancing detection accuracy and real-time tracking performance.
- Proposed and implemented a fall prevention posture detection framework based on Spatial-Temporal Graph Convolutional Networks (ST-GCN).

eSports Gaze Behavior Detection | CNN, DETR, Few-Shot Learning

Feb 2022-April 2023

- Designed and developed a personalized eSports assistant using deep learning for gaze tracking across video frames.
- Built a cross-frame gaze estimation model using convolutional neural networks and temporal feature alignment.
- Proposed a novel few-shot learning framework allowing for rapid adaptation to users with minimal calibration data.
- Final system supports real-time inference and operates with <50ms latency on standard GPUs.

Health Management System with Health Code and Trip Card | SQL, C++, Data Structure

June 2020-May 2021

- Developed a health code system, including QR code generation, data management, and epidemic risk level updates.
- Designed a structured user database supporting registration, trip tracking, and automated status updates.
- Implemented modular system features such as location-based risk evaluation and dynamic color coding.

EXPERIENCE

Research Intern at BrainML Lab, Georgia Tech | **GNN, Multiple Agents Behavior Analysis, Neural Science**

Jan-Dec 2025

Topic: Trajectory analysis of animal behaviors using graph neural networks and geodesic loss optimization.

- Conducted research on machine learning methods for trajectory analysis to understand animal behavior patterns.
- Implemented algorithms to predict mice movement and modeled spatiotemporal interactions using graph structures.
- Developed training pipelines and introduced a novel geodesic loss function that improved performance.

Research Assistant at HKUST, Hong Kong | **Stable Diffusion, Image Segmentation**

Jan-June 2024

Topic: Enhancing segmentation from segmented datasets containing noisy labels using generative models

- Conducted research on applying noisy label segmentation to diffusion models for image generation tasks.
- Proposed a novel approach for learning from noise-infused labels in both unsupervised and semi-supervised settings.
- Utilized unlabeled and noisy-labeled image datasets to improve segmentation performance using Stable Diffusion.

Research Assistant at CUHK, Hong Kong | **Diffusion Models, OCT Imaging**

Dec 2022-Mar 2023

Topic: Multiplicative speckle noise denoising of OCT medical images based on Diffusion Model

- Developed a DDPM-based denoising method for speckle noise in OCT medical images to improve diagnostic accuracy.
- Implemented noise modeling, diffusion training pipelines, and enhanced convergence speed and prediction efficiency.
- Reproduced baseline models including DN-CNN, U-Net, BM3D, TCFL-OCT, and DDPM for comparative evaluation.
- Wrote Python scripts to compute denoising metrics such as PSNR, SSIM, and GCMSE.

Participant at CVPR 2023 BDD100K Challenges | Object Tracking, Autonomous Driving, DETR, YOLO-NAS May-June 2023

Topic: Multi Object track to guide autonomous driving through machine learning algorithms

- Implemented object detection with advanced detectors including YOLO-NAS, YOLOv8, and DINO.
- Developed a multi-object tracking pipeline for autonomous driving using state-of-the-art machine learning models.
- Fine-tuned models on the MOT20 dataset and applied ByteTrack and BoT-SORT for object tracking.
- Ranked in the top 20% for Multi-Object Tracking (MOT) and achieved first place in Object Detection (OD) tasks.

Winter School at North Carolina State University, Raleigh, NC | **BCI, LSTM, Transformer, EEG**

Dec 2022-Feb 2023

Topic: Motion classification of brain-computer interface signals using machine learning algorithms

- Conducted research on motion classification of brain-computer interface (BCI) signals using machine learning methods.
- Developed EALNet, a custom BCI classification model combining **EEGNet, LSTM, and Transformer** architecture.
- Achieved 10% improvement in classification accuracy on standard BCI Competition datasets.

Undergrad Research Assistant at HUST, Wuhan, China | **OCR, MMOCR, DBNet**

Sep-Dec 2022

Topic: Recognition and extraction of musical notes using Optical Character Recognition (OCR)

- Conducted research on text detection of musical notation using Optical Character Recognition (OCR) techniques.
- Built and annotated a specialized dataset from online musical tune data for model training and evaluation.
- Evaluated models including **DBNet, DBNet++, and PANet** using the MMOCR framework and detection libraries.